Analysing Sepal and Petal Patterns Across Iris Species

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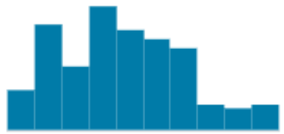
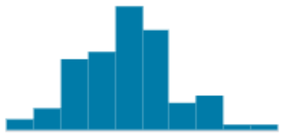
Dataset Selection

Iris.csv (5.11 kB)

Detail | Compact | Column

About this file

SQLite database containing the same data as Iris.csv

Id SPL-SPW-PTL-PTW(CM)	# SepalLengthCm Length of the sepal (in cm)	# SepalWidthCm Width of the sepal (in cm)	# PetalLengthCm Length of the petal (in cm)	# PetalWidthCm Width of the petal (in cm)	Species Species name
 1 150	 4.3 7.9	 2 4.4	 1 6.9	 0.1 2.5	**3** unique values
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa

Dataset Overview

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("Iris.csv")
```

```
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
```

Dataset Overview

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
df.shape
```

```
(150, 6)
```

```
df.isnull().sum()
```

```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

```
df.duplicated().sum()
```

```
0
```

With the describe() function, we can see that there are 150 rows in this dataset.
The average sepal length is 5.84 cm.
The minimum sepal width is 2.0 cm.
About 75% of iris flowers in this dataset have a petal length lower than 5.1 cm.

corr() Error: Non-Numeric Data

```
df.corr()
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[6], line 1
----> 1 df.corr()

File /opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages/pandas/core/frame.py:10054, in DataFrame.corr(self, method, min_periods, numeric_only)
  10052 cols = data.columns
  10053 idx = cols.copy()
> 10054 mat = data.to_numpy(dtype=float, na_value=np.nan, copy=False)
  10056 if method == "pearson":
  10057     correl = libalgos.nancorr(mat, minp=min_periods)

File /opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages/pandas/core/frame.py:1838, in DataFrame.to_numpy(self, dtype, copy, na_value)
   1836 if dtype is not None:
   1837     dtype = np.dtype(dtype)
-> 1838 result = self._mgr.as_array(dtype=dtype, copy=copy, na_value=na_value)
   1839 if result.dtype is not dtype:
   1840     result = np.array(result, dtype=dtype, copy=False)

File /opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages/pandas/core/internals/managers.py:1732, in BlockMana
   1730         arr.flags.writeable = False
   1731 else:
-> 1732     arr = self._interleave(dtype=dtype, na_value=na_value)
   1733     # The underlying data was copied within _interleave, so no need
   1734     # to further copy if copy=True or setting na_value
   1736 if na_value is not lib.no_default:

File /opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages/pandas/core/internals/managers.py:1794, in BlockMana
   1792     else:
   1793         arr = blk.get_values(dtype)
-> 1794     result[rl.indexer] = arr
   1795     itemmask[rl.indexer] = 1
   1797 if not itemmask.all():

ValueError: could not convert string to float: 'Iris-setosa'
```

Fix Code

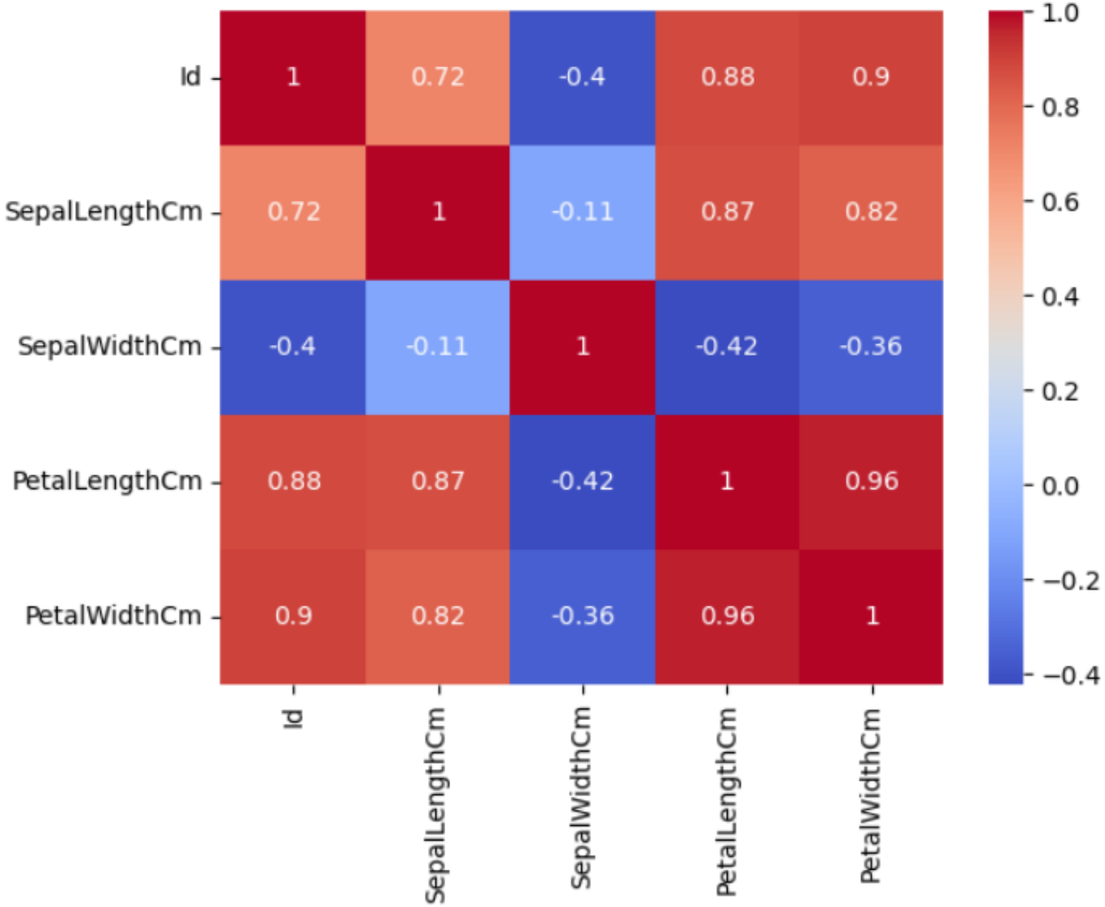
```
df.corr(numeric_only=True)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Id	1.000000	0.716676	-0.397729	0.882747	0.899759
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754	0.817954
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516	-0.356544
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000	0.962757
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757	1.000000

Using the numeric_only parameter, this function calculates correlations only for numeric columns, ignoring non-numeric data to prevent errors

Correlation Heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```

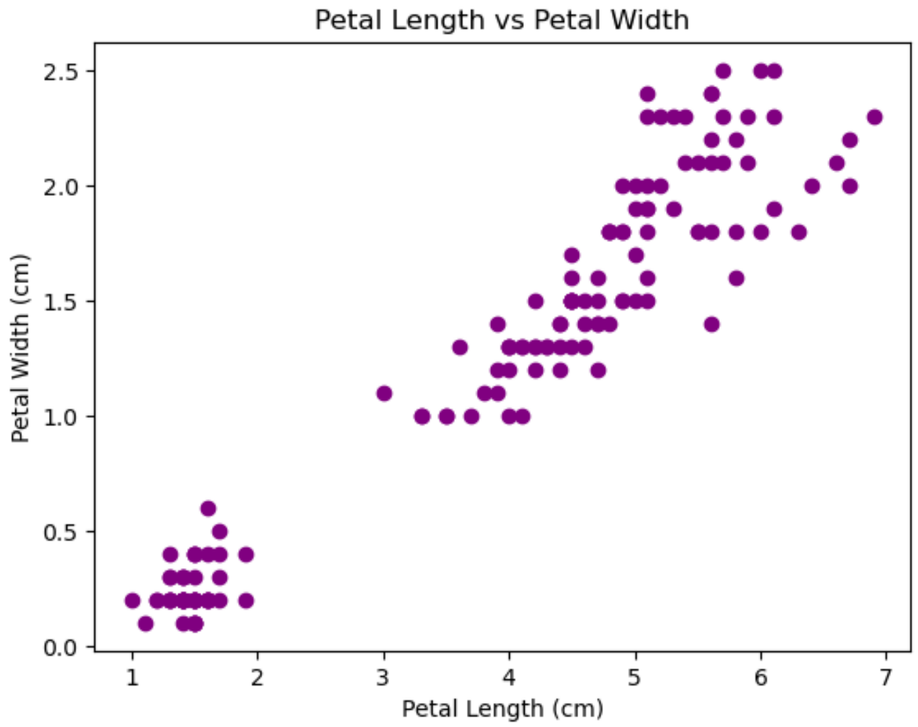


We are now using Seaborn to display a heatmap of the correlations between numeric columns in our Iris dataset.

Because PetalLength and PetalWidth are very similar, we keep only one to simplify our analysis and reduce redundancy.

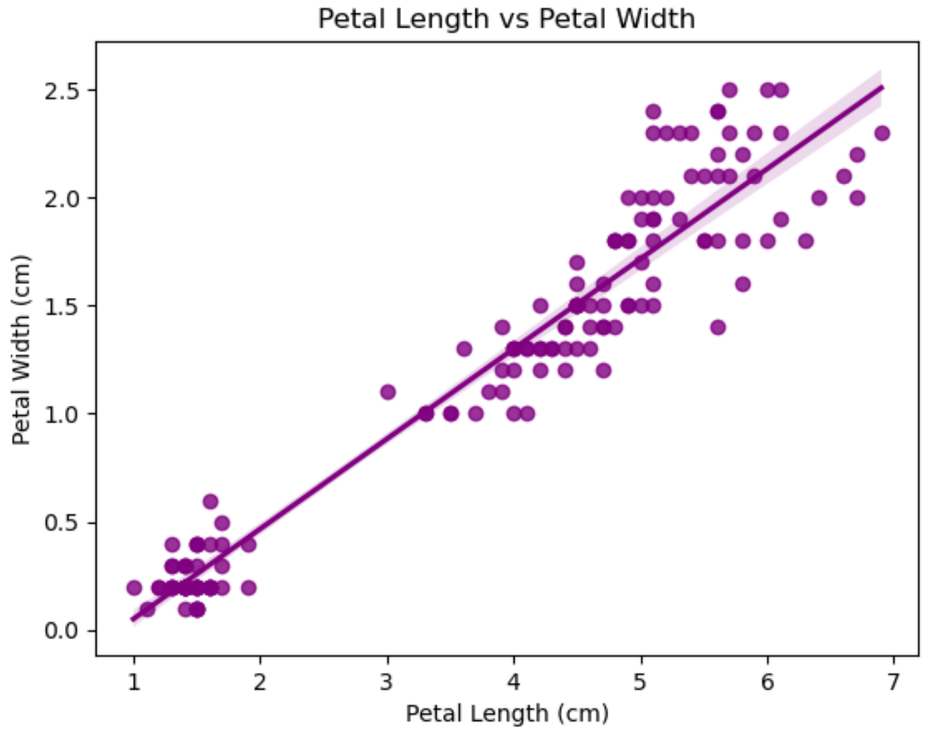
Positive Linear Relationship

```
plt.plot(df['PetalLengthCm'], df['PetalWidthCm'], 'o', color='purple')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Petal Width (cm)')
plt.title('Petal Length vs Petal Width')
plt.show()
```




```
sns.regplot(x='PetalLengthCm', y='PetalWidthCm', data=df, color='purple')
plt.xlabel('Petal Length (cm)')
plt.ylabel('Petal Width (cm)')
plt.title('Petal Length vs Petal Width')
```

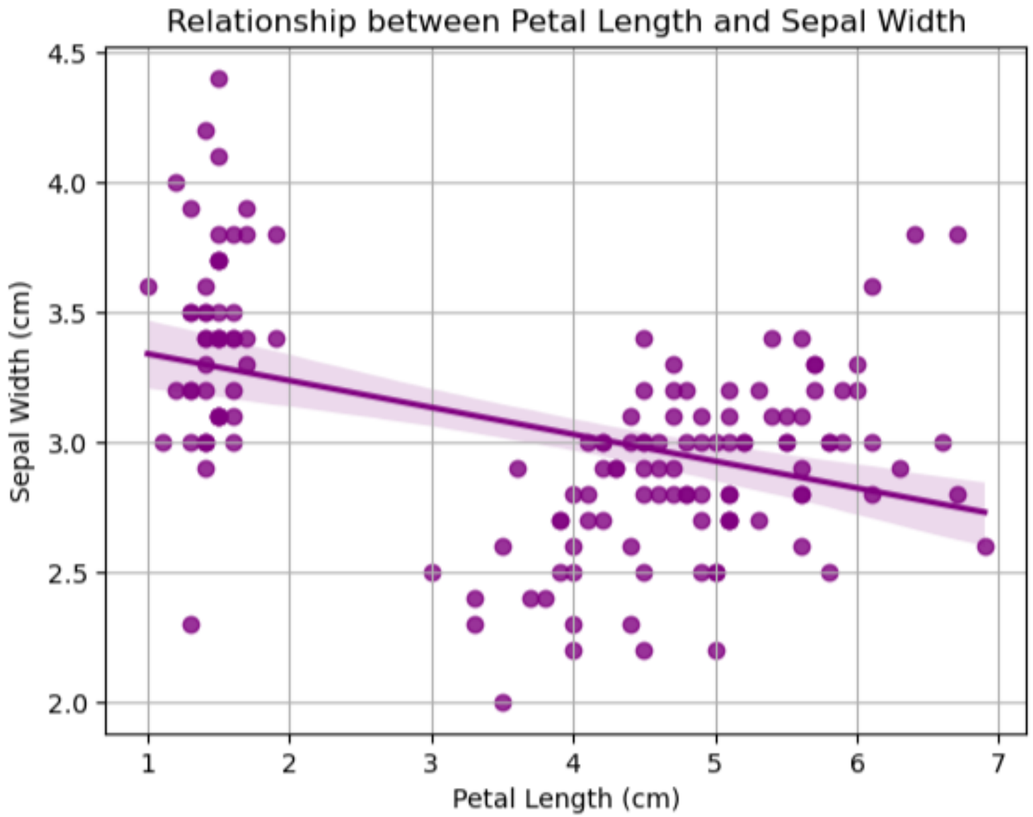
```
Text(0.5, 1.0, 'Petal Length vs Petal Width')
```



The scatter plot of PetalLength vs. PetalWidth suggests a linear relationship. The regression line confirms that larger PetalLength values correspond to larger PetalWidth values

Negative Correlation

```
sns.regplot(x='PetalLengthCm', y='SepalWidthCm', data=df, color='purple')
plt.title("Relationship between Petal Length and Sepal Width")
plt.xlabel("Petal Length (cm)")
plt.ylabel("Sepal Width (cm)")
plt.grid(True)
plt.show()
```



Similarly, the scatter plot of PetalLength versus SepalWidth reveals a negative correlation, indicating that the longer the petal, the narrower the sepal tends to be

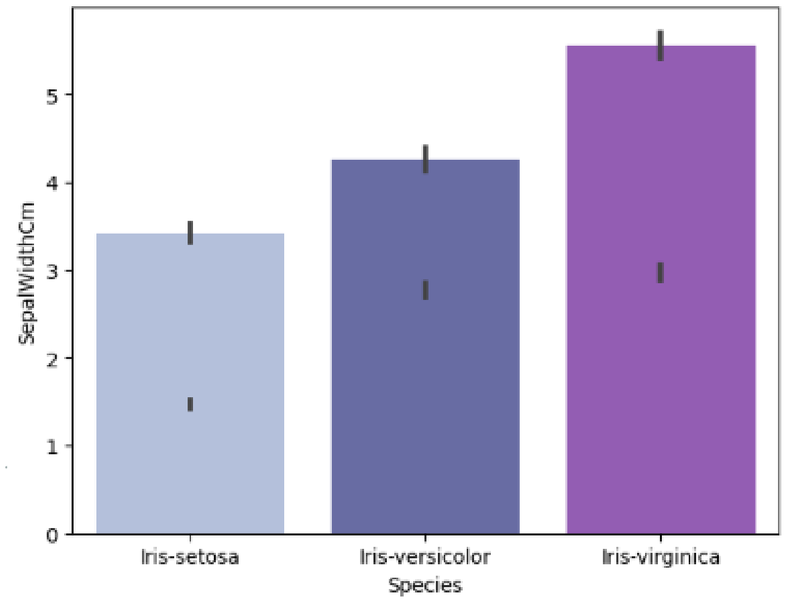
Species Feature Overview

```
df.groupby('Species')[['PetalLengthCm', 'SepalWidthCm']].mean()
```

Species	PetalLengthCm	SepalWidthCm
Iris-setosa	1.464	3.418
Iris-versicolor	4.260	2.770
Iris-virginica	5.552	2.974

```
sns.barplot(x='Species', y='PetalLengthCm', data=df,  palette=['#adbee2', '#5e63ad', '#964fc1'])
sns.barplot(x='Species', y='SepalWidthCm', data=df, palette=['#adbee2', '#5e63ad', '#964fc1'])
```

```
<Axes: xlabel='Species', ylabel='SepalWidthCm'>
```



By grouping the data by species, we can see that PetalLength and SepalWidth vary across Iris types, with Iris-setosa having the shortest petals and widest sepals, while the other species have longer petals and narrower sepals



Thank You